DNS · MULTIPHASE FLOWS · HPC & GPU COMPUTING

Akhil S. Pillai

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RESEARCH INTERESTS

Direct numerical simulation of compressible gas–liquid flows; six-equation two-fluid modelling; bluff-body wakes and vortex dynamics; GPU-accelerated scientific computing on national HPC systems.

RESEARCH EXPERIENCE

Project Associate
Jul 2025 – present

Department of Aerospace Engineering, IIT Madras — Prof. A. Sameen's group

- DNS of turbulent air–water flow past a circular cylinder (Re = 100–300) using an in-house Fortran six-equation two-fluid solver.
- Production runs on 8–30 million-cell unstructured meshes, distributed across GPUs on PARAM Shakti (NSM) with one GPU per MPI rank.
- Validating computed shedding frequencies against published Strouhal-number data; studying wake vortex dynamics and droplet–air interface behaviour.

Research Intern
Dec 2024 – Jun 2025

Department of Aerospace Engineering, IIT Madras

- Built and ran compressible multiphase simulations with a six-equation multifluid formulation (HLLC-Batten interfacial fluxes; ~4 million tetrahedral cells); validated against the Yeom & Chang (2013) benchmark to within about 1%.
- Resolved vortex shedding and interface dynamics around a cylinder at low-to-moderate Reynolds numbers; mesh generation in ICEM CFD; batch workflows on PARAM Shakti.

Summer Fellow
May – Jul 2024

IIT Madras Summer Fellowship Programme

- Machine learning for aerodynamic prediction: trained a CNN on XFoil-generated data to predict lift and drag coefficients from airfoil images, with ANN and PINN baselines in PyTorch (CUDA).

EDUCATION

B.Tech, Aerospace Engineering
2021 – 2025

Karunya Institute of Technology and Sciences, Coimbatore

- Thesis: *Numerical simulation of flow over a bluff body using a compressible multifluid six-equation formulation* — carried out at IIT Madras (host: Prof. A. Sameen); guide: Dr. Aldin Justin Sundaraj.

INDUSTRY EXPERIENCE

Internship Trainee
Jun 2023

Hindustan Aeronautics Limited, Bengaluru

- LCA Tejas production line: component fabrication, structural assembly, and system integration in a live defence-aviation environment.

Earlier internships
2021 – 2022

- Omspace Rocket & Exploration (launch vehicles, 2022); Brahmastra Aerospace Systems (industrial aerodynamics, 2022; aircraft vehicle design, 2021–22); XUANTA Developers (quantum computing, 2022); student delegate, 7th Bangalore Space Expo (CII, 2022).

TECHNICAL SKILLS

COMPUTING	Fortran, MPI, GPU computing, Python / PyTorch (CUDA), LaTeX
SIMULATION	DNS, six-equation two-fluid modelling, Riemann solvers (HLLC-Batten), ICEM CFD, ANSYS
DESIGN	SolidWorks

MEMBERSHIP

Student member, Aeronautical Society of India (2023–present).